W-C-1684
April 4, 1977

FEDERAL SPECIFICATION

CASSETTES, MAGNETIC TAPE, AUDIO RECORDING
(TWIN-HUB, COPLANAR)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for twin-hub, coplanar, tape cassettes, with unrecorded magnetic recording audio tape suitable for mono/stereo compatible use.

1.2 Classification. Each individual configuration of cassette covered by this specification shall be identified by an item identifying part number, consisting of indicators compiled from the part number codes (see 1.2.1) as shown below:

C-120	-	AP	-	I	-	WP
Playing time indicator (see 1.2.1.1)		Oxide characteristic indicator (see 1.2.1.2)		Packaging indicator (see 1.2.1.3)		Leader indicator (see 1.2.1.4)

1.2.1 Part number codes.

1.2.1.1 Playing time indicator. The playing time indicator indicates the minimum total playing time (both sides) at 4.76 cm/s (1-7/8 in/s). The playing time shall be indicated as follows:

Indicator code	Playing time per side at 4.76 cm/s (1-7/8 in/s)
C-30	15 minutes
C-45	22.5 minutes
C-60	30 minutes
C-90	45 minutes
C-120	60 minutes

1.2.1.2 Oxide characteristic indicator. The oxide characteristic indicator identifies a particular oxide by its characteristics, as specified herein. The oxide characteristic shall be indicated as follows:

Indicator code	Oxide characteristic
AP	See table III

1.2.1.3 Packaging indicator. The packaging indicator indicates whether the cassette is packaged in a unit container or packaged for bulk packing (see 5.1). Packaging shall be indicated as follows:

Indicator code	Packaging
I	Method I - Unit packaging
II	Method II - Bulk packaging

1.2.1.4 Leader indicator. The indicator "WP" identifies cassettes supplied without leaders or with magnetic oxide coated leaders. The absence of the indicator identifies cassettes supplied with uncoated leaders.

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Test Method Std. No. 406 - Plastics: Methods of Testing.
Fed. Test Method Std. No. 406/1013 - Tensile Properties of Thin Sheets and Films.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

International Electrotechnical Commission (IEC) Recommendations:

IEC 94 and Amendment No. 1 - Magnetic Tape Recording and Reproducing Systems: Dimensions and Characteristics.
IEC 94A - Supplement to Publication 94: Cassette for Commercial Tape Records and Domestic Use, Dimensions and Characteristics.

(Copies may be obtained from the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

Electronic Industries Association (EIA) Standards:

RS-399A - Dimensional Standard Coplanar Magnetic Tape Cartridge Type CP II (Compact Cassette).

(Copies may be obtained from the Electronic Industries Association, Engineering Department, 2001 Eye Street, NW, Washington, DC 20006.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

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Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

American Society for Testing and Materials (ASTM) Standard:

D-638 - Tensile Properties of Plastics: Method of Test for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 General. The magnetic tape and the cassettes furnished under this specification shall be new and shall have been manufactured in accordance with the requirements given in IEC 94A; the dimensional measurements given in EIA RS-399A; and as specified herein. Whenever this specification differs from IEC 94A and EIA RS-399A then this specification shall govern.

3.1.1 Component parts. One complete unit shall consist of a length of magnetic oxide-coated tape, with or without tape leaders, as applicable, contained within a cassette housing as specified herein, and packaged in accordance with 5.1.1 or 5.1.2 as applicable.

3.2 Qualification. The items furnished under this specification shall be products that have been tested, and passed the qualification tests specified herein, and have been listed or approved for listing on the applicable Federal Qualified Products List (QPL) (see 4.3 and 6.3). The general rules and regulations governing Federal Qualified Products Lists, as covered by the "Provisions Governing Applications by Manufacturers for Inclusion on Federal Qualified Products Lists" (Appendix 4-A to GSA Federal Standardization Handbook FPMR 101-29), shall apply (see 6.4).

3.2.1 Process change. Qualification, once established, applies only to those cassettes manufactured by the same process used to make the cassettes that passed the qualification tests. All proposed process changes shall be reported to the General Services Administration, Federal Supply Service, with a statement by the supplier as to the general nature of the changes and the extent of such changes on the delivered product. The General Services Administration reserves the right to require six weeks notice, and sufficient samples of the new product, for quality assurance tests, from the supplier prior to delivery of products manufactured under any such change in process. The foregoing requirement does not in any way relieve the manufacturer of delivery requirements, quality control, and testing necessary to insure that all products delivered under this specification are equal to those products originally submitted for qualification.

3.3 Materials.

3.3.1 Tape. The tape shall consist of a layer of magnetic material on a backing of polyester plastic, or equivalent, and shall have the physical, magnetic, and other characteristics specified herein.

3.3.2 Cassettes. The cassettes shall be constructed of materials suitable for the intended use and shall conform to the physical, dimensional, and other requirements specified herein.

3.4 Cassette dimensional requirements. The cassette dimensions shall conform to the measurements given in EIA RS-399A.

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3.5 Cassette physical requirements.

3.5.1 Cassette assembly. Cassette halves shall be bound together with an ultrasonic weld or with screws. The finish of the cassette halves shall be clean, and free from cracks and sharp flash. The seam on the front of the cassette housing where the two cassette halves come together shall be located outside the normal tape path.

3.5.1.1 Drop resistance. The cassette shall remain bound together with no dislocation of parts within the overall tolerances given in EIA RS-399A and must perform satisfactorily in the reference recorder/reproducer after being subjected to the drop test of 4.6.8.4.

3.5.2 Corner tape guides. Corner tape guides shall be provided and shall have a minimum hub diameter of 7.11 mm (0.280 in). Rotating corner tape guides shall be flanged.

3.5.3 Windows. Each side of the cassettes shall be provided with a window to observe the amount of wound and unwound tape. Minimum viewable dimensional measurements shall be 6.35 mm (0.25 in) wide and 19.05 mm (0.75 in) long. Each window-pane shall be capable of withstanding a pressure of 1.59 kg (3.5 lbs), evenly distributed over the minimum allowable window surface, for a period of not less than 5 seconds, without cracking, loosening, or showing visible distortion. There shall be no visible indication of excess adhesive around the window-pane edges.

3.5.4 Friction liners. The cassettes shall be provided with antifriction liners of suitable material. The liners shall be of uniform workmanship, free from manufacturing defects, such as, frays, tears, and wrinkles. The tape contact surface of the liner shall be electrically conductive to reduce build up of static charges to the tape. The antifriction area of each liner shall cover not less than 90 percent of the tape contact surface, excluding the cassette window area.

3.5.5 Hubs. The hub mold-injection-point shall not be on any part of the hub where it can be in contact with the tape and friction liners. The hub shall be constructed in such a way that no bulging of the tape occurs when the tape is wound around the hub.

3.5.6 Pressure pad assembly. The pressure pad assembly shall consist of a device to apply pressure between the tape and record/reproduce head and shall meet the requirement of 3.8.3.

3.5.7 Magnetic shield. Each cassette shall be provided with a hum reducing magnetic shield positioned directly behind the pressure pad.

3.5.8 Labels. Each cassette shall be provided with two labels positioned within the label areas given in EIA RS-399A. Labels shall securely adhere to the housing and shall not loosen or curl when subjected to the environmental conditions of 3.11. Labels shall be marked with the playing time indicator (e.g., C-60), playing time per side at 4.76 cm/s (1-7/8 in/s) and the manufacturers name. Each label shall have a lined or unlined writing space of 7.75 square centimeters (1.0 square inch) minimum. Printing ink used on labels shall not smear or come off when subjected to the environmental conditions stated herein. Labels shall readily accept lead pencil and ballpoint pen markings.

3.6 Tape dimensional measurements.

3.6.1 Length. The length of tape contained within a particular type cassette shall be in accordance with table I.

3.6.2 Width. The tape width shall be $3.81 \begin{smallmatrix} + 0.00 \\ - 0.05 \end{smallmatrix}$ mm ($0.150 \begin{smallmatrix} + 0.000 \\ - 0.002 \end{smallmatrix}$ in).

3.6.3 Thickness. The overall thickness of the tape (base and coating) contained within a particular type cassette shall be in accordance with table I.

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TABLE I. Tape length and thickness

Playing Time Indicator	Tape Length		Tape Thickness (max.)	
	meters	feet	mm	in
C-30	42.90 min. - 47.55 max.	141 min. - 156 max.	0.020	0.0008
C-45	64.30 min. - 68.90 max.	211 min. - 226 max.	0.020	0.0008
C-60	85.80 min. - 90.55 max.	282 min. - 297 max.	0.020	0.0008
C-90	128.60 min. - 133.20 max.	422 min. - 437 max.	0.015	0.0006
C-120	171.50 min. - 176.20 max.	563 min. - 578 max.	0.010	0.0004

3.7 Tape leader. This specification covers cassettes supplied with uncoated leaders, and cassettes supplied either without leaders or with magnetic oxide coated leaders. Unless otherwise specified (see 6.1), cassettes shall be supplied with an uncoated polyester, or equivalent, leader attached to both ends of the tape. The width of the tape leader shall be in accordance with the tape width requirement of 3.6.2.

3.7.1 Leader length. Each leader shall be 30.48 cm (12 in) minimum to 45.72 cm (18 in) maximum in length, with a nominal thickness of 0.038 mm (0.0015 in).

3.7.2 Leader-to-hub and tape-to-hub connections. Leader-to-hub and tape-to-hub connections shall be capable of withstanding a minimum force of 1 kilogram (2.2 pounds) for 10 seconds.

3.7.3 Leader splices. Tape-to-leader splices shall be capable of withstanding a force of 227 grams (8 ounces) for five minutes, without an increase of the gap between tape and leader of more than 0.127 mm (0.005 in). Before testing, the gap between the tape and leader shall not exceed 0.635 mm (0.025 in). No overlap is permitted.

3.7.3.1 Splicing tape width. The splicing tape shall not protrude beyond the edges of the magnetic tape and tape leader.

3.7.3.2 Splicing tape adhesive. The splicing tape adhesive shall not ooze after the complete cassette has been subjected to any one of the temperature and humidity conditions given in 3.11.

3.8 Mechanical requirements.

3.8.1 Maximum friction torque of both hubs. The friction torque of both hubs, measured in the cassette itself at the nearly full hub, shall not exceed 2.7×10^{-3} Nm (27 gcm), when measured in accordance with 4.6.6.1.

3.8.2 Maximum friction torque of both hubs with hold back torque. With a hold back torque of 0.8×10^{-3} Nm (8 gcm) applied to the nearly empty hub, the required torque to be applied to the nearly full hub, when moving tape, shall not exceed 5.5×10^{-3} Nm (55 gcm), when measured in accordance with 4.6.6.1.

3.8.3 Pressure pad pressure. When the record/reproduce head is engaged into the cassette, in accordance with the maximum dimension given in EIA RS-399A, the pressure of the pad on the magnetic head over the head to pad contact area shall be 0.005 to 0.015 N/mm^2 (0.5 to 1.5 g/mm^2).

3.9 Tape physical requirements.

3.9.1 Yield strength. The 1 percent offset yield point; the force corresponding to 3 percent elongation; and the tensile force at break, for the tape of a particular type cassette, when measured in accordance with 4.6.13.1, shall equal or exceed the value specified in table II for that particular type cassette.

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3.9.2 Elongation under stress. The elongation of the tape for a particular type cassette, when measured in accordance with 4.6.13.2, shall not exceed the value specified in table II for that particular type cassette.

TABLE II. Magnetic tape physical requirements

Characteristic	Requirement (3.81 mm [0.150 in] tape width)		
	C-30; C-45; C-60	C-90	C-120
Yield strength Elongation under stress	0.544 Kg (1.2 lbs) 0.5 percent	0.363 Kg (0.8 lbs) 0.5 percent	0.272 Kg (0.6 lbs) 0.5 percent

3.9.3 Anchorage. The magnetic layer of the tape shall show no visible evidence of anchorage failure after completion of all tests. Anchorage failure may be manifested as a separation of the magnetic coating from the base material of the tape, or internal failure of the coating layer itself.

3.9.4 Splices. All magnetic tape lengths shall be continuous and splice free.

3.9.5 Layer-to-layer adhesion. The adhesion characteristics of the tape shall be such that the frictional torque shall not exceed the value given in 3.8.2 after the complete cassette has been subjected to any of the temperature and humidity conditions given in 3.11.

3.10 Magnetic tape performance requirements.

3.10.1 Maximum recording level. The input voltage level of a 333 Hz signal, with operating bias current (see 4.6.7.1.3), necessary to produce 3 percent third harmonic distortion in the output, when measured in accordance with 4.6.7.4.1, relative to the reference recording level (see 4.6.7.1.4), shall fall within the applicable limits specified in table III.

3.10.2 Output at maximum recording level. The output voltage level of a 333 Hz signal, recorded for 3 percent third harmonic distortion in the output, with operating bias current (see 4.6.7.1.3), when measured in accordance with 4.6.7.4.1, relative to the reference output level (see 4.6.7.1.5), shall fall within the applicable limits specified in table III.

3.10.3 Sensitivity. The output voltage level of a 333 Hz signal, recorded at standard recording level (see 4.6.7.1.6) with operating bias current (see 4.6.7.1.3), at any point on the tape, when measured in accordance with 4.6.7.4.2, relative to the reference output level (see 4.6.7.1.5) shall fall within the applicable limits specified in table III.

3.10.4 Harmonic distortion. The third harmonic distortion at standard recording level (see 4.6.7.1.6), when measured in accordance with 4.6.7.4.2 shall not exceed the applicable maximum value specified in table III.

3.10.5 Frequency response. The output voltage levels, at frequencies of 6000 Hz, 8000 Hz, and 10000 Hz, when measured in accordance with 4.6.7.4.3, and when normalized to the output voltage level at 333 Hz and compared with the response of the "Reference Medium" section of the reference tape (see 4.6.7.1.1), shall fall within the applicable limits specified in table III.

3.10.6 Uniformity. The uniformity of the output signal level at the test frequencies of 333 Hz and 8000 Hz, when measured in accordance with 4.6.7.4.4, shall be such that the maximum deviation from the maximum peak output shall not exceed the applicable value in table III.

3.10.7 Reference level to bias noise ratio. The reference level to bias noise ratio, when measured in accordance with 4.6.7.4.5, shall equal or exceed the applicable value specified in table III.

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TABLE III. Magnetic tape performance requirements

Characteristic	Measurement: Oxide Characteristic AP			Units
	C-30, C-45 C-60	C-90	C-120	
Maximum recording level relative to reference recording level	+ 3 max. - 0 min.	+ 3 max. - 1 min.	+ 3 max. - 2 min.	Decibels
Output at maximum recording level relative to reference output level	+ 3 max. - 0 min.	+ 3 max. - 1 min.	+ 3 max. - 2 min.	Decibels
Harmonic distortion (max.) @ standard recording level	1	1	1	Percent
Sensitivity	- 18 max. - 20 min.	- 18 max. - 21 min.	- 18 max. - 23 min.	Decibels
Frequency response:				
@ 6000 Hz	± 2	± 2	+ 2, - 4	Decibels
@ 8000 Hz	± 2, - 3	± 2, - 3	+ 2, - 5	Decibels
@ 10000 Hz	± 3	+ 3, - 4	+ 2, - 6	Decibels
Uniformity				
@ 333 Hz	20	20	20	Percent
@ 8000 Hz	30	30	30	Percent
Reference level to bias noise ratio	50	50	50	Decibels

3.11 Environmental extremes. When tested in accordance with 4.6.8, cassettes shall not warp and shall conform to the mechanical requirements in 3.8.2 and the tape magnetic performance requirements in 3.10.3, 3.10.5, and 3.10.6 after being subjected to any of the following temperature and humidity conditions: 65°C to 70°C (149°F to 158°F) at 0 to 30 percent relative humidity for 24 hours; -30°C to -25°C (-22°F to -13°F) for 24 hours; and 45°C to 50°C (113°F to 122°F) at a relative humidity of 85 to 95 percent for 24 hours.

3.12 Short term operation. When tested in accordance with 4.6.9.1, the cassettes shall perform in accordance with the requirements specified herein.

3.13 Operational life. After the cassette has been subjected to the operational life test of 4.6.9.2, the cassette shall perform in accordance with the requirements specified herein, and shall conform to the mechanical requirement of 3.8.2, and the average output level of a 333 Hz signal, recorded at standard recording level (see 4.6.7.1.6), shall not decrease by more than 2 dB.

3.14 Workmanship and general examination. The tape and tape cassettes shall be manufactured and processed in a careful and workmanlike manner in accordance with good practice. Both the magnetic oxide surface and base of the tape shall be free from any raised areas, dust, flakes, powder, holes, foreign matter, scratches, creases and other defects which would render the tape unsuitable for its intended use. The condition of tape slitting, the cassette, and the packaging containers shall be examined for workmanship and cleanliness.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection of end item (see 4.4).
(c) Acceptance inspection (see 4.5).
(d) Inspection of preparation for delivery (see 4.7).

4.3 Qualification inspection. Qualification inspection shall be conducted at a laboratory designated by the General Services Administration, Federal Supply Service. Qualification inspection shall consist of each of the examinations, measurements, and tests listed in table IV, and shall be performed in the order given in 4.6.1 and figure 1. All samples subjected to the examinations, measurements, and tests specified in table IV must conform to the associated requirements specified in Section 3.

4.3.1 Qualification sample size. Except as otherwise specified in 4.3.2, the qualification sample shall consist of eight cassettes of the type (e.g., C-120; see 1.2) for which qualification is desired, packaged in accordance with 5.1.1.

4.3.2 A qualification sample for C-60 cassettes, with a particular tape base, tape thickness, tape oxide characteristic, and cassette housing, shall constitute a qualification sample for C-30 and C-45 cassettes using the same particular tape base, tape thickness, tape oxide characteristic, and cassette housing.

4.4 Quality conformance inspection of end items. The contractor shall select a sample number of units from each lot offered for delivery. Sample units shall be selected in accordance with 4.4.1 and the contractor shall subject the sample units to the appropriate examinations, measurements, and tests of table IV to determine compliance with the specification requirements. All necessary test equipment and facilities required to conduct the subject examinations and tests shall be furnished by the contractor.

TABLE IV. Examinations, measurements, and tests

Examinations, measurements, and tests	Requirement paragraph	Test method paragraph
GROUP A		
General examination	3.5.1, 3.5.3, 3.5.6, 3.5.7, 3.5.8, 3.14, and 5.1	4.6.4
General dimensional measurements	3.4	4.6.5.1
Magnetic tape length	3.6.1 and 3.9.4	4.6.12.1
Leader length	3.7.1	4.6.12.3
GROUP B		
Short term operation	3.12	4.6.9.1
Sensitivity	3.10.3	4.6.7.4.2
Frequency response	3.10.5	4.6.7.4.3
Uniformity	3.10.6	4.6.7.4.4
Magnetic tape width	3.6.2	4.6.12.2
Anchorage	3.9.3	4.6.14

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TABLE IV. Examinations, measurements, and tests (cont.)

Examinations, measurements, and tests	Requirement paragraph	Test method paragraph
GROUP C		
Frictional torque	3.8.1 and 3.8.2	4.6.6.1
Pressure pad pressure	3.8.3	4.6.6.2
Splice gap	3.7.3	4.6.12.4
Splice strength	3.7.3	4.6.12.4
Splice tape width	3.7.3.1	4.6.13.4
Leader-to-hub or tape-to-hub strength	3.7.2	4.6.13.3
Drop test	3.5.1.1	4.6.8.4
GROUP D		
Detail dimensional measurements	3.4	4.6.5.2
Maximum recording level	3.10.1	4.6.7.4.1
Output at maximum recording level	3.10.2	4.6.7.4.1
Harmonic distortion	3.10.4	4.6.7.4.2
Reference level to bias noise ratio	3.10.7	4.6.7.4.5
Cassette physical examination and measurement	3.5.4, 3.5.5, 3.5.2, 3.5.3 and 3.5.8	4.6.10 and 4.6.11
Magnetic tape yield strength	3.9.1	4.6.13.1
Magnetic tape elongation under stress	3.9.2	4.6.13.2
Environmental conditions	3.11	4.6.8
Operational life	3.13	4.6.9.2

4.4.1 Sampling for quality conformance inspection. Sampling for quality conformance inspection of end items shall be in accordance with MIL-STD-105 and as specified in 4.4.1.1, 4.4.1.2, 4.4.1.3, and 4.4.1.4.

4.4.1.1 Sampling for group A examinations, measurements, and tests. Sampling for group A examinations, measurements, and tests shall be in accordance with MIL-STD-105, Special Inspection Level S-2, at an acceptable quality level (AQL) of 10 percent defective.

4.4.1.2 Sampling for group B examinations, measurements, and tests. Sampling for group B examinations, measurements, and tests shall be in accordance with MIL-STD-105, Special Inspection Level S-2, at an acceptable quality level (AQL) of 10 percent defective.

4.4.1.3 Sampling for group C examinations, measurements, and tests. Sampling for group C examinations, measurements, and tests shall be in accordance with MIL-STD-105, Special Inspection Level S-1, at an acceptable quality level (AQL) of 4 percent defective.

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4.4.1.4 Sampling for group D examinations, measurements, and tests. The group D examinations, measurements, and tests shall be required initially and as often as deemed necessary to assure continued compliance with the specified requirements. Four sample cassettes shall be subjected to the group D examinations, measurements, and tests given in table IV. Failure to conform to any requirement of the group D examinations, measurements, and tests shall be cause for the rejection of the cassettes represented by the sample.

4.5 Acceptance inspection. Lots offered for delivery shall be subjected to an acceptance inspection by the Government consisting of the Group A and B examinations, measurements, and tests given in table IV. Acceptance tests on shipment lots shall be conducted by the qualifying activity.

4.5.1 Sampling for acceptance inspection. Sampling for acceptance inspection shall be in accordance with 4.4.1.1 and 4.4.1.2.

4.6. Examinations, measurements, and tests.

4.6.1 Order of qualification inspection. Qualification inspection shall be performed in the following order in accordance with figure 1.

(a) All cassettes under test shall be subjected to the examinations, measurements, and tests for general examination, general dimensions, frictional torque, pressure pad pressure, maximum recording level, output at maximum recording level, sensitivity, harmonic distortion, frequency response, and reference level to bias noise ratio, listed in table IV.

(b) The cassettes shall then be divided into four groups. Cassettes in groups 1, 2, and 3 shall be subjected to the environmental conditions of 4.6.8.1, 4.6.8.2, and 4.6.8.3 respectively. Cassettes in group 4 shall be subjected to the drop test of 4.6.8.4. All cassettes, groups 1 thru 4 shall then be subjected to the mechanical measurement of 4.6.6.1 for conformance to the frictional torque requirement of 3.8.2. Cassettes in groups 1, 2, and 3 shall then be subjected to the magnetic tape measurements for sensitivity, frequency response, and uniformity listed in table IV. Cassettes in groups 1, 2, and 3 shall be examined for warpage, loose or curled labels, and printing ink that will smear or come off the labels.

(c) One cassette from each group shall then be subjected to the operational life test of 4.6.9.2. Measurement for conformance to the frictional torque requirement shall be for the requirement of 3.8.2 only.

(d) One cassette from each group shall then be subjected to the examinations, measurements, and tests for physical examination and measurements, anchorage, tape length, tape width, leader length, splice gap, splicing tape width, leader-to-hub or tape-to-hub strength, splice strength, magnetic tape yield strength, and magnetic tape elongation under stress listed in table IV.

4.6.2 Preliminary conditioning. Test units shall be subjected to the test environment of 4.6.3 for a period of not less than 24 hours prior to performance of any measurement or test specified herein.

4.6.3 Test environment. Unless otherwise specified herein, all measurements and tests shall be performed at an ambient temperature of 21.1°C to 26.7°C (70°F to 80°F) and a relative humidity of between 45 and 55 percent.

4.6.4 General examinations. Test units shall be examined for conformance to the applicable requirements of 3.5.1, 3.5.3, 3.5.6, 3.5.7, 3.5.8, 3.14, and 5.1.

4.6.5 Cassette dimensional measurements.

4.6.5.1 General dimensional measurements. The following listed dimensions shall be measured for conformance to the measurements given in EIA RS-399A. Appropriate "GO", "NO-GO" gauges may be used to determine conformance to outside dimensional requirements. The record/reproduce head opening width, the erase head opening width, and the ends of head opening (see [f], [g], and [h] below) shall be measured at upper and lower corners.

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Outside dimensions to be measured:

(a) Length.
(b) Width.
(c) Thickness.
(d) Capstan holes; size and location.
(e) Locating holes; size and location.
(f) Erase head opening; height and width.
(g) Record/reproduce head opening; height and width.
(h) Extreme ends of head openings.
(i) Hub spindle holes; size and location.
(j) Hub spoke inside diameter.
(k) Hub spoke; size, number, and location.
(l) Support planes flatness and parallelism.

4.6.5.2 Detail dimensional measurements. All other dimensions shall be measured to determine conformance with 3.4.

4.6.6 Cassette mechanical measurements.

4.6.6.1 Frictional torque. A torque tester, such as the Information Terminals Model M-400, or Minnetech Labs Model MTM 200 series, or equivalent, shall be used to perform this test. The cassette shall be inserted into the tester such that the nearly full hub is on the takeup side. The torque tester shall be started, and with no holdback torque applied to the supply hub, the average torque reading on the meter shall be noted for conformance to the requirement of 3.8.1. The foregoing test shall be repeated with a holdback torque of 6.8×10^{-3} Nm (8 gm. cm) applied to the supply hub to determine conformance to the requirement of 3.8.2. The tape shall then be wound on the opposite hub and the foregoing tests repeated.

4.6.6.2 Pressure pad pressure. The pressure pad pressure shall be measured for conformance to the requirement of 3.8.3.

4.6.7 Magnetic tape performance measurements.

4.6.7.1 Definitions.

4.6.7.1.1 Reference tape and cassette. The reference tape shall consist of a recorded and unrecorded length of DIN Calibration Tape 4.75/3.81 mm in accordance with DIN 45 513, Sheet 6, in a cassette housing that conforms to IEC 94A (see 6.2).

4.6.7.1.2 Secondary reference tape. The secondary reference tape is an unrecorded length of tape, the magnetic characteristics of which have been calibrated against those of the "Reference Medium" section of the reference tape. The secondary reference tape is used for preliminary adjustments of the reference test recorder/reproducer and as a working reference.

4.6.7.1.3 Operating bias current. Operating bias current is that bias current through the recording head which will give a 2.5 dB fall-off (over bias peak) of the peak output from the "Reference Medium" section of the reference tape (see 4.6.7.1.1) when a 6.3 kHz signal is recorded at standard recording level (see 4.6.7.1.6).

4.6.7.1.4 Reference recording level. The reference recording level is that voltage level of a 333 Hz input signal which, in conjunction with operating bias current (see 4.6.7.1.3), will produce a permanent induction in the "Reference Medium" section of the reference tape (see 4.6.7.1.1), such that upon playback the output signal level conforms to the reference output level.

4.6.7.1.5 Reference output level. The reference output level is the output level that is obtained by reproducing the "Reference Level" section of the DIN Calibration Tape (see 4.6.7.1.1).

4.6.7.1.6 Standard recording level. Standard recording level is 20 dB below the reference recording level (see 4.6.7.1.4).

4.6.7.1.7 Standard output level. Standard output level is that reproduce output voltage measured on playback of a 333 Hz signal that has been recorded on the "Reference Medium" section of the reference tape (see 4.6.7.1.1) at standard recording level (see 4.6.7.1.6) with operating bias current (see 4.6.7.1.3).

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4.6.7.2 Reference test recorder/reproducer. For all the measurements specified under 4.6.7.4, a reference test recorder/reproducer in proper adjustment shall be used. The reference test recorder/reproducer shall have the following minimum features.

(a) The recorder/reproducer shall be a stereophonic cassette type and except as otherwise specified herein shall be in accordance with the applicable requirements of IEC 94A.

(b) The tape speed shall be 4.76 cm/s (1-7/8 in/s) ± 1 percent.

(c) The harmonic distortion introduced by the record/reproduce amplifiers and the record/reproduce head shall not exceed 0.5 percent at all levels and frequencies used in the magnetic tape performance measurements.

(d) The overall signal-to-noise ratio shall be 55 dB or higher. The overall signal-to-noise ratio is defined as the ratio of the reference output level to the total unweighted playback noise after erasure of a 333 Hz signal recorded at reference recording level and in the absence of a new signal. Thus, bias and erase noises are included as well as the playback and record amplifier noises. All frequencies between 40 Hz and 12 kHz shall be measured.

(e) The flutter shall be less than 0.15 percent rms NAB Weighted at a tape speed of 4.76 cm/s (1-7/8 in/s).

(f) The record/reproduce head(s) shall be a two channel quarter track head(s). The recording head gap length shall be 200 microinches nominal.

(g) The erase signal frequency shall be 80 kHz or higher. The signal strength shall be such as to reduce the level of a signal recorded at reference recording level on the "Reference Medium" section of the reference tape at least 60 dB.

(h) The bias frequency shall be 80 kHz or higher and adjustable in amplitude.

(i) The reproducing characteristic shall be that which gives a flat response, within the limits given in figure 2, when reproducing a track recorded with the recorded tape flux characteristics stated in Clause 7 of IEC 94 (see IEC Publication 94, Amendment 1) for a tape speed of 4.76 cm/s (1-7/8 in/s).

(j) The recording characteristic shall be such that the overall record/reproduce frequency response is flat, within the limits given in figure 2, using the "Reference Medium" section of the reference tape (see 4.6.7.1.1) with operating bias current (see 4.6.7.1.3).

(k) Any commercial recorder which conforms to the requirements specified herein, may be used for the applicable tests and measurements specified herein.

(l) For the qualification tests and the lot acceptance tests specified herein, the General Services Administration will use a Nakamichi 1000, manufactured by Nakamichi Research, Inc., or equivalent, audio cassette recorder/-reproducer.

4.6.7.2.1 Short term operation and operational life test apparatus. For the short term operation and the operational life tests specified herein, the General Services Administration will cycle the cassette on a Califone model AV-15, or equivalent, cassette recorder/reproducer.

4.6.7.3 Preliminary adjustment of test recorder/reproducer, and determination of reference signal levels. Before commencing the magnetic tape performance measurements, the reference test recorder/reproducer shall be set up and adjusted, and operating bias current and reference signal levels determined in accordance with the following, for both channel 1 and channel 2:

(a) For all measurements, the reference test recorder/reproducer shall be terminated with its proper load impedance.

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(b) All test signals shall be sinewave signals.

(c) The recorder/reproducer shall be thoroughly cleaned and the record/reproduce head demagnetized.

(d) The record/reproduce head shall be adjusted for correct azimuthal orientation.

(e) Determine the reference output level using the "Reference Level" section of the reference tape (see 4.6.7.1.1).

(f) Determine the reference recording level, the operating bias current, the standard recording level, and the standard output level using the "Reference Medium" section of the reference tape.

(g) Determine that the reproduce frequency response is flat, within the limits given in figure 2, using a tape recorded with the recorded tape flux characteristics stated in clause 7 of IEC 94 for a tape speed of 4.76 cm/s (1-7/8 in/s).

(h) Determine that the overall frequency response is flat, within the limits given in figure 2, using the "Reference Medium" section of the reference tape with operating bias current.

4.6.7.4 Test procedures. The following magnetic tape performance measurements are applicable to all tracks.

4.6.7.4.1 Maximum recording level measurement. A 333 Hz signal shall be recorded onto the cassette tape to be tested, at that voltage level which, in conjunction with operating bias current (see 4.6.7.1.3), will produce a permanent induction in the tape such that upon playback the output signal will have 3 percent third harmonic distortion. The input and output voltages, at maximum recording level, shall be measured and noted. The input voltage level, at the maximum recording level, shall be compared with the reference recording level (see 4.6.7.1.4) to determine conformance to 3.10.1. The output voltage level, at the maximum recording level, shall be compared with the reference output level (see 4.6.7.1.5) to determine conformance to 3.10.2.

4.6.7.4.2 Sensitivity measurement. A 333 Hz signal shall be recorded onto the cassette tape to be tested at standard recording level (see 4.6.7.1.6) with operating bias current (see 4.6.7.1.3). The recorded signal shall then be played back and the output voltage measured and noted. Correction shall be made for the response of the reference recorder/reproducer with a secondary reference tape at operating bias current, when a secondary reference tape is used. The output voltage level from the tape under test, at standard recording level, shall be compared with the reference output level (see 4.6.7.1.5) to determine conformance to 3.10.2. Third harmonic distortion in the output shall be measured to determine conformance to 3.10.4.

4.6.7.4.3 Frequency response measurements. Signals of 333 Hz, 6000 Hz, 8000 Hz, and 10000 Hz shall be recorded onto both the cassette tape to be tested and the "Reference Medium" section of the reference tape (see 4.6.7.1.1) at 25 dB below reference recording level (see 4.6.7.1.4), with operating bias current (see 4.6.7.1.3). The reference tape shall be played back and the output voltage at each frequency measured and noted. Correction shall be made for the response of the reference recorder/reproducer with a secondary reference tape at operating bias current, when a secondary reference tape is used. The outputs of the reference tape, at 6000 Hz, 8000 Hz, and 10000 Hz, shall be normalized to the output of the reference tape at 333 Hz and expressed in dB as the response of the reference tape. The tape under test shall be played back and the output voltage at each frequency measured and noted. The outputs of the tape under test, at 6000 Hz, 8000 Hz, and 10000 Hz, shall be normalized to the output of the tape under test at 333 Hz and expressed in dB as the response of the tape under test. The response of the tape under test shall be compared to the response of the reference tape to determine conformance to 3.10.5.

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4.6.7.4.4 Uniformity measurement. Two 3-minute (minimum) segments, consisting of one 333 Hz segment and one 8000 Hz segment, shall be recorded onto the cassette tape to be tested at 10 dB below reference recording level (see 4.6.7.1.4) with operating bias current (see 4.6.7.1.3). The recorded signals shall then be played back and the output voltage variations measured by means of a graphic level recording system having a flat frequency response from dc to 1 or 2 Hz and having an upper frequency limit (6 dB down) of approximately 6 Hz. The tape uniformity shall be measured by determining the peak variations in output as a percentage of maximum peak output over any 10 second interval for conformance to 3.10.6. All nonrecurring pips shall be disregarded.

4.6.7.4.5 Reference level to bias noise ratio. The cassette tape to be tested shall be externally erased. A segment of the tape shall then be recorded with operating bias only (input gain controls set to minimum, no input signal). This segment of tape shall then be played back and the output voltage measured (in dB) at the output of a NAB weighted filter. This value shall be taken as the bias noise level. Determine the reference level to bias noise ratio as the value of the reference output level minus the value of the bias noise level for conformance to 3.10.7.

4.6.8 Environmental tests. Cassettes selected in accordance with 4.6.1(b) shall be subjected to the high temperature, low temperature, and humidity conditions of 4.6.8.1, 4.6.8.2, and 4.6.8.3 and shall then be retested for conformance to the mechanical requirements of 3.8.2. All cassettes shall then be tested to the requirements of 3.10.3, 3.10.5, and 3.10.6.

4.6.8.1 High temperature. The cassettes under test shall be subjected to 65°C to 70°C (149°F to 158°F) and a relative humidity of between 0 and 30 percent for a period of 24 hours. The cassettes shall be brought back to "test environment" (see 4.6.3) at a temperature gradient of no greater than 7.2°C (45°F) per hour and shall remain at "test environment" at least two hours before further tests.

4.6.8.2 Low temperature. The cassettes under test shall be subjected to -30°C to -25°C (-22°F to -13°F) for a period of 24 hours. The cassettes shall be brought back to "test environment" (see 4.6.3) at a temperature gradient of no greater than 7.2°C (45°F) per hour and shall remain at "test environment" at least two hours before further tests.

4.6.8.3 Humidity. The cassettes under test shall be subjected to 45°C to 50°C (113°F to 122°F) and a relative humidity of between 85 and 95 percent for a period of 24 hours. The cassette shall be brought back to "test environment" (see 4.6.3) by opening the test chamber door and allowing the cassette temperature to stabilize to the "test environment" and remain at the "test environment" for at least two hours before further test.

4.6.8.4 Drop test. The drop test shall consist of four drops from a height of 30 inches onto a concrete floor covered with asphalt tile. The first two drops shall be on alternate ends and the second two drops shall be on alternate faces. Upon completion of the drops, the cassette shall be inspected for conformance to the requirement of 3.5.1.1 and tested for conformance to the requirement of 3.8.2.

4.6.9 Operational tests.

4.6.9.1 Short term operation test. The cassette to be tested shall be subjected to one slow cycle and one fast cycle, in that order, to determine conformance to 3.12. A slow cycle shall consist of playing the tape from end to end in one direction, then turning the cassette over and playing the tape from end to end in the other direction. A fast cycle shall consist of fast rewinding the tape from end to end in one direction, then turning the cassette over and fast rewinding the tape from end to end in the other direction.

4.6.9.2 Operational life test. A 333 Hz signal shall be recorded onto tracks 1 and 2 of the cassette tape to be tested at standard recording level (see 4.6.7.1.6) with operating bias current (see 4.6.7.1.3). The recorded signals shall be played back and the average output voltage for each track measured and noted as the initial output level for that track. The test cassette shall then be life tested by repetitive cycling until it has reached 28 cycles or fails, whichever comes first.

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One cycle shall consist of playing the cassette from end to end in one direction, and then fast rewinding from end to end in the other direction. The cassettes shall be turned over after every 7 cycles. Upon completion of the 28 cycles the cassettes shall be retested for conformance to the mechanical requirements of 3.8.2 and the previously recorded signals shall be played back and the average output voltage, for each track measured (in dB) and noted as the final output level for that track. The initial output voltage level for each track shall be compared with the final output voltage level for each track for conformance to 3.13.

4.6.10 Corner tape guides, window dimensions, and label writing space measurements. The cassette corner tape guide hub diameter, the window dimensions, and the label writing space shall be measured for conformance to the applicable requirements in 3.5.2, 3.5.3, and 3.5.8.

4.6.11 Cassette physical examination. The cassette shall be visually examined for conformance to the applicable requirements in 3.5.4 and 3.5.5.

4.6.12 Magnetic tape and leader dimensions measurements.

4.6.12.1 Tape length. The tape length shall be measured for conformance to the requirement of 3.6.1. The tape shall be visually examined for conformance to the requirement of 3.9.4.

4.6.12.2 Tape width. The tape width shall be measured for conformance to the requirement of 3.6.2.

4.6.12.3 Leader length. The tape leader length shall be measured for conformance to the requirement of 3.7.1.

4.6.12.4 Splice gap and splicing tape width. The splice gap shall be measured for conformance to the requirement of 3.7.3. The splicing tape width shall be visually examined for conformance to the requirement of 3.7.3.1.

4.6.13 Magnetic tape and leader physical measurements.

4.6.13.1 Tape yield strength.

4.6.13.1.1 Apparatus. The apparatus for determining the specified tensile properties shall be as described in paragraph 3.2 of Fed. Test Method Std. No. 406, Method 1013.

4.6.13.1.2 Preparation of samples. Samples shall be conditioned as specified in paragraph 4.6.2.

4.6.13.1.3 Procedure. A strip of tape not less than 7 inches in length shall be clamped in the grips set for an initial separation of 4 inches. The test specimen shall be clamped in the testing machine taking care to align the long axis of the specimen with an imaginary line joining the points of attachment of the grips to the machine. The grips shall be tightened evenly and firmly to the degree necessary to prevent slipping of the specimen during the test. The rate of grip separation shall be 2 inches per minute. Not less than five specimens of a particular type of tape shall be tested. A force strain diagram shall be obtained. Means shall be provided for calibration of the strain axis. The 1.0 percent offset yield strength shall be determined as described in ASTM Standard D-638. The tangent (approximating the initial slope) to the force-strain curve shall be obtained by drawing a straight line between the origin and the point on the curve corresponding to one-half the specified force (see table II). If the force-strain curve between these two points deviates by more than 0.1 percent elongation from the straight line drawn, a new straight line shall be drawn from the origin to one quarter of the specified force. The 1.0 percent offset yield strength determined in this manner and the force corresponding to 3 percent elongation shall equal or exceed the value specified in table II for the particular type of tape being evaluated. In the event that the breakage of a particular tape occurs prior to reaching the 1.0 percent offset yield strength or 3 percent elongation, the force at break shall equal or exceed the value specified in table II.

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4.6.13.2 Tape elongation under stress.

4.6.13.2.1 Apparatus. The apparatus for this test shall consist of standard means of measuring length, positive nonslipping clamping devices, a 25 gram weight, and weights as specified in 4.6.13.2.3.

4.6.13.2.2 Preparation of samples. Samples shall be prepared as specified in 4.6.2. Samples at least 610 mm (24 in) long shall be clamped so as to hang in the test area for at least 24 hours under no externally applied stress before tests are begun. No less than five specimens of a particular type of tape shall be tested.

4.6.13.2.3 Procedure. Before any weight is hung on the test samples, a mark shall be made approximately 508 mm (20 in) from the point of clamping to measure the elongation and recovery of the sample. The distance between the mark and clamp point shall be measured accurately to the nearest 0.254 mm (0.01 in) with an applied tension of approximately 25 grams. This distance shall be taken as the base distance for calculation of elongation. When the measurement of the base distance has been made, the test shall begin. A test weight of 340 grams for tape from C-30, C-45, and C-60 cassettes; 185 grams for tape from C-90 cassettes; and 120 grams for tape from C-120 cassettes; shall be attached to the tape below the mark at zero time and allowed to hang undisturbed for 180 minutes ± 30 seconds, at which time the weight shall be removed from the tape. The tape shall be allowed to hang under its own weight for an additional 180 minutes ± 30 seconds. The distance between the mark and the point of clamping shall then be measured to the nearest 0.254 mm (0.01 in) with an applied tension of approximately 25 grams. The difference between the base distance and the final distance shall be expressed as a percent of the base distance to determine compliance with 3.9.2.

4.6.13.3 Leader-to-hub and tape-to-hub strength. The leader-to-hub and tape-to-hub strength shall be measured for conformance to the requirement of 3.7.2.

4.6.13.4 Splice strength. The splice strength shall be measured for conformance to the requirement of 3.7.3.

4.6.14 Anchorage. A visual examination shall be made of the tape after completion of all tests to determine conformance to 3.9.3.

4.7 Inspection of preparation for delivery. An inspection shall be made to determine whether the preservation, packaging, packing and marking comply with the requirements in Section 5. Defects shall be scored in accordance with table V. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery and selected at random just prior to closing and sealing. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

TABLE V. Classification of preparation for delivery defects

Examine	Defects
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Material	Any component missing or damaged or wrong type.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling, or distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be in accordance with 5.1.1 or 5.1.2, as specified (see 6.1).

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5.1.1 Method I unit packaging. Each cassette shall be completely enclosed in a close fitting unit container of rigid plastic, as shown in figure 3. The container shall be hinged, self-locking and shall support the cassette in a manner which will prevent crushing or distortion. At least one of the unit containers largest faces shall be transparent to permit viewing the cassette without opening the container. Each cassette, or the unit container, shall be provided with hub locks that will prevent rotation of the hubs during shipment and handling. Each unit container, with cassette inside, shall be wrapped in either a clear heat sealable plastic bag, a clear plastic shrink wrapper, or a clear cellophane wrapper, to form a dust-proof package.

5.1.2 Method II, bulk packaging. Bulk packaging shall be in accordance with the supplier's commercial practice, which shall include enclosing cassettes, either individually or in groups, in either a clear heat sealable plastic bag, a clear plastic shrink wrapper, or a clear cellophane wrapper, to form a neat dust proof package. Cassettes shall be provided with hub locks that will prevent rotation of the hubs during shipment and handling.

5.2 Intermediate packaging. Cassettes packaged in accordance with 5.1.1, or individually bulk packaged (see 5.1.2), shall be packaged in an intermediate package in accordance with the supplier's normal commercial practice.

5.3 Packing. Packaged cassettes (see 5.1.1, 5.1.2, and 5.2) shall be packed in a shipping container in accordance with the supplier's normal commercial practice to ensure safe delivery at destination, and shall be acceptable by common carrier under National Motor Freight Classification or Uniform Freight Classification.

5.4 Marking. Each shipping container shall be marked in accordance with Fed. Std. No. 123. Marking shall include an item identifying part number (see 1.2).

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Cassette item identifying part number (see 1.2).
(c) Quantity required.
(d) Inspection responsibility (see 4.1).
(e) Specify packaging required (see 5.1 and 1.2.1.3).

6.2 Reference cassette. DIN Calibration Reference Cassettes may be obtained from Order Department, BASF Systems, Crosby Drive, Bedford, MA 01730, Part No. 70220.

6.3 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time for opening of bids, been tested and approved for inclusion on Qualified Products List QPL-W-C-1604, whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the General Services Administration, Federal Supply Service, Washington, DC 20406, and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification, etc." (see 6.4).

6.4 Copies of "Provisions Governing Qualification, etc." may be obtained upon application to General Services Administration, Federal Supply Service, Office of Standards and Quality Control, Washington, DC 20406.

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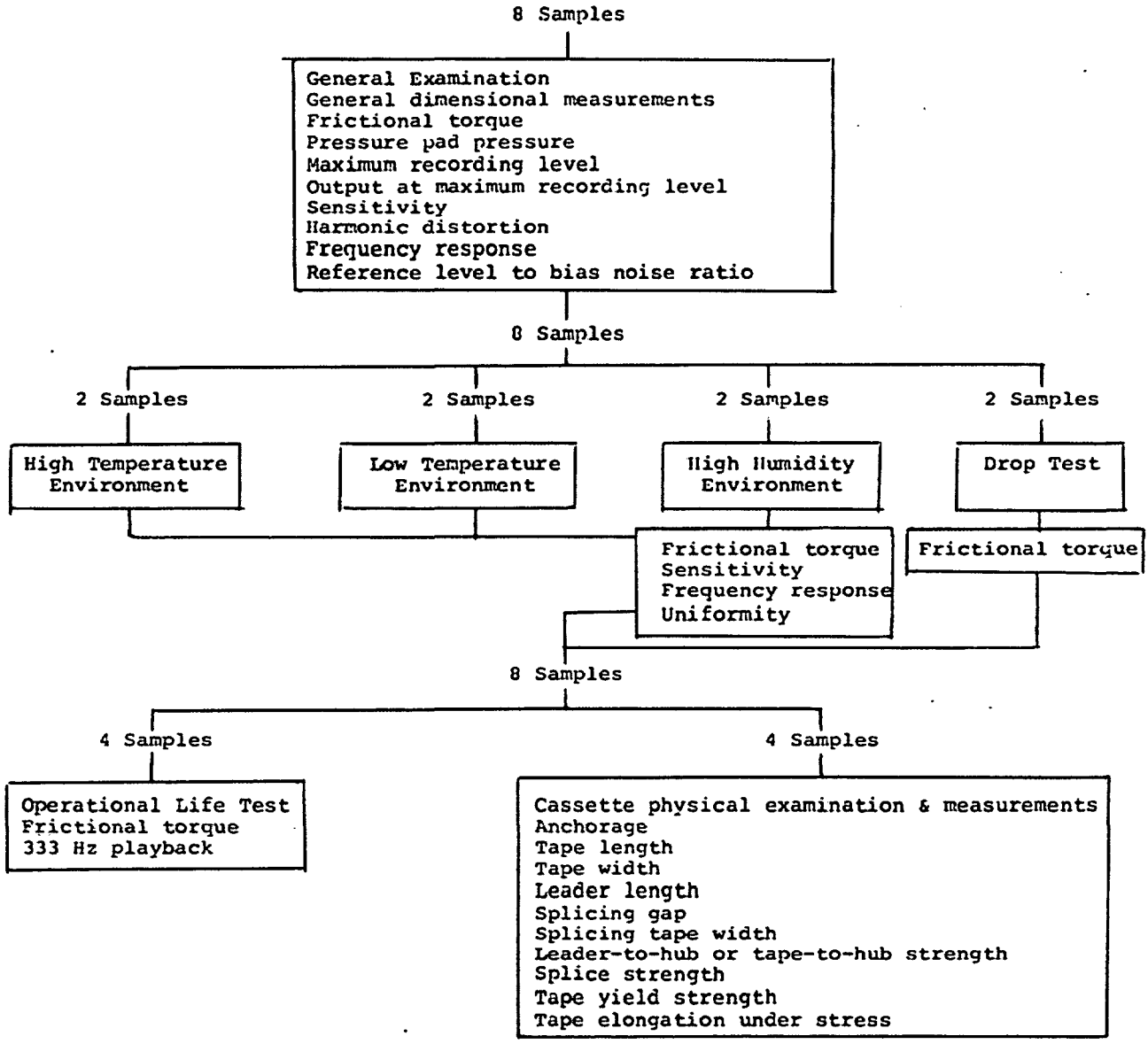


FIGURE 1

Qualification Inspection Flow Chart

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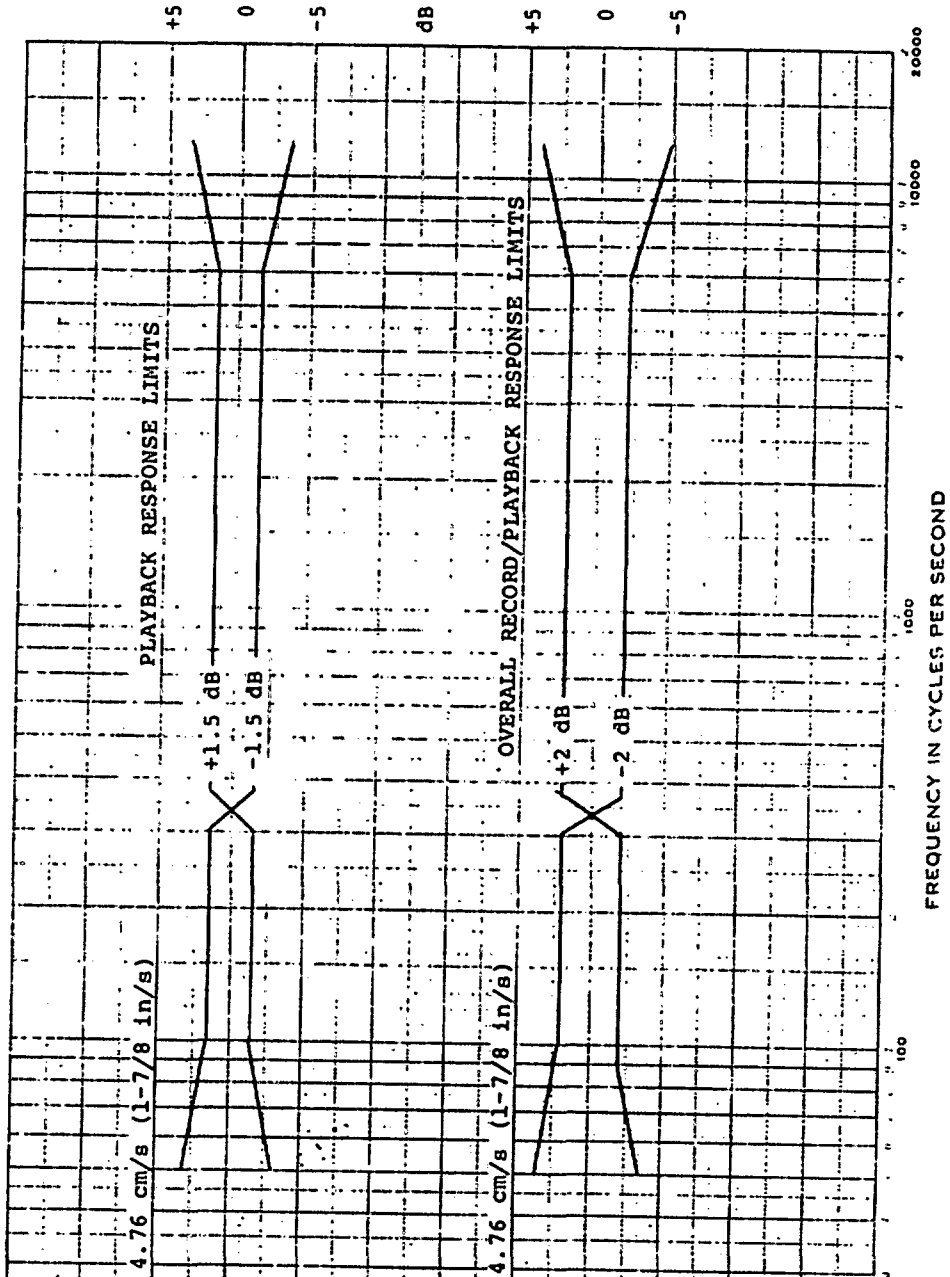


FIGURE 2
FREQUENCY RESPONSE LIMITS

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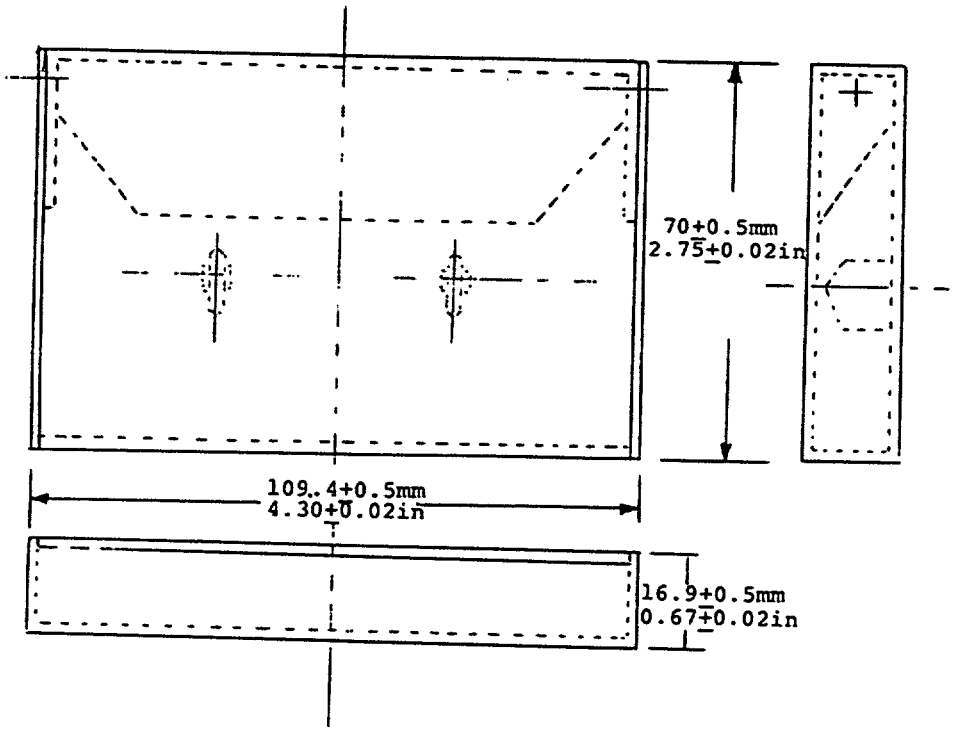


FIGURE 3. PLASTIC CONTAINER

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